

Internship certificate

11.07.2025

This is to certify that Ms. Paula Luisa Brüdrieh completed her internship at Third Element Aviation GmbH, Bielefeld from May 12, 2025 until July 11, 2025 in the field of Hardware Development.

Third Element Aviation GmbH is a commercial developer and manufacturer of unmanned aerial vehicle systems (UAS). Its core products include highly automated industrial drones. The company currently employs around 26 people and is headquartered in Bielefeld.

During her internship, Ms. Brüdrieh was entrusted with the following tasks and projects:

Hardware Development

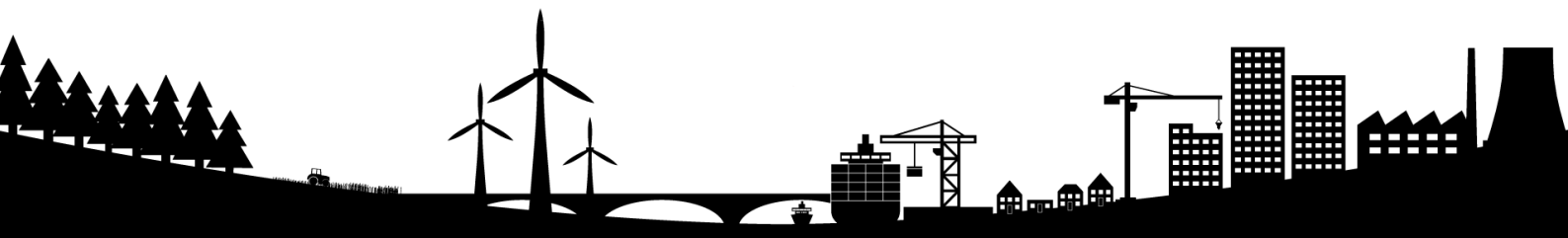
- Main project: Assisted in the design, prototyping, and testing of a new “FSB” (Flight Support Board).
 - Contributed to the layout design of a 6-layer PCB with nearly 400 components using Fusion.
 - Gained knowledge of component selection, Fusion libraries, and routing, particularly concerning internal data layers.
 - Conducted research on signal integrity and applied Fusion’s impedance matching extension to improve data transmission.
 - Assembled prototype boards and performed soldering with a reflow oven.
 - Supported the initial load tests and helped identify potential areas for improvement.
- Side project: Independently designed the layout and routing for a small “PWM” PCB.
- Performed diverse soldering tasks with both reflow oven and soldering iron.

Production

- Disassembled a test hexacopter for hardware replacement and reassembled the system.
- Acquired insights into the full workflow from assembly to maiden flight (calibration, motor tests, etc.).
- Contributed to miscellaneous tasks in the production department, such as assembling new parachute tensioning tools.

Other Activities

- Received an introduction to Jira, participated in sprint planning, and contributed to documentation maintenance.
- Attended a VTOL flight demonstration for clients.



Ms. Brüdriich proved to be a motivated and quick learner who approached all her assignments with diligence, precision, and enthusiasm. She demonstrated strong analytical skills, technical understanding, and commitment to her tasks. Her ability to adapt to new tools and workflows quickly made her a valuable member of our team.

Behavior and Conduct

Her behavior towards supervisors, colleagues, and external partners was always exemplary at all times. Ms. Brüdriich consistently distinguished herself through her high reliability, excellent team spirit, and respectful, professional communication. She contributed in every respect to a positive and productive working atmosphere.

We would like to thank Ms. Brüdriich for her excellent contribution during her internship at Third Element Aviation GmbH. We regret her departure and would have been pleased to continue working with her. We wish her every success and all the best for both her academic path and future professional career.

Third Element Aviation GmbH



Benjamin Wiens

